Homework - 2

CSL-100/MAL-400 Introduction To Programming

Question 1

A number n is called left truncatable prime if n and all numbers obtained by successively removing its left most digits are prime. (Similarly right truncatable prime is defined)
Ex 313 is a left truncatable prime 313 is prime, 13 is prime, 3 is prime.
313 is also right truncatable 313 is prime, 31 is prime, 3 is prime
Write a C programme which takes a number n as input and then tells whether it is left truncatable, right truncatable or both.

Question 2

Create a function named rev to reverse a number. Use this to create a function named differev that will return the difference of the reversed number from the original number (d=N-R). For example, for N=514 reversed number is 415 and the difference is 99. Verify that differev(N) + rev(N) = N

Question 3

In number theory, a polite number is a positive integer that can be written as the sum of two or more consecutive positive integers. Rest of the integers are said to be impolite. For e.g. 7 and 18 are a polite numbers as 7 = 3+4 and 18=5+6+7. The first few polite numbers are
3, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 17, 18, ...
The politeness of a positive integer is the number of ways in which we can represent the given number as sum of positive consecutive integers. For e.g. 9 has a politeness of 2 as 9 = 2+3+4 as well as 4+5. Write a function, politeFunc, which takes a positive integer and prints various ways in which it can be represented as a sum of consecutive positive integers. If the number is impolite, output the string "I am RUDE!".

Input Format
Enter the number.

Constraints

Each possible representation is printed in a new line.
Output Format
The possible combinations or "I am RUDE!".

Sample Input 0
9
Sample Output 0
2 3 4
4 5

Question 4

An inversion in an array of real numbers A is any pair (i,j) such that i¡j and A[i]¿A[j]. For e.g. in the array [25,7,51,3], the inversions would be (0,1),(0,3),(1,3) and (2,3). You have to write a program which takes 7 distinct non-negative integers from the user as array entries. The program then should print out the inversions in the given array-each separated by a tab character. If there are no inversions in the given array (i.e. array is sorted), your program should output "No inversions here!"

Question 5

An inversion vector, I(A) of an array A is an array of n-1 elements. The ith entry of the array I(A) is the number of elements greater than A[i] preceding it in the original array A. Clearly the first entry of I(A) is always zero as no element in an array precedes its first element. If the array is sorted then all elements of inversion vector would be zero.
For e.g. in the array [25,7,52,3,21], the inversion vector would be [0,1,0,3,2]. You have to write a program which takes 7 distinct non-negative integers. Store these numbers in an array and compute the inversion vector of the given array. The program must also print each entry of the inversion array separated by a tab character.

Question 6

Watson gives Sherlock an integer array A of size n. His challenge is to find if there is an element in the array such that the sum of all elements to its left is equal to the sum of all elements to its right. Write a function findElement(int A[ ], n) that takes Array A and integer n as input and return 1 if such element exists else return 0.

Input Format
Enter the number of elements in the array and create the array.

Output Format
0 or 1

Sample Input 0
4
3 1 2 3
Sample Output 0
0
Explanation : No such element is there.

Sample Input 1
5
4 11 2 6 0
Sample Output 1
1
Explanation 2 4 11 6 0 in this order sum of element right of 11 is same as the sum to the left.

Question 7

Two friends like to pool their money and go to the ice cream parlor. They always choose two distinct flavors and they try to spend all of their money. Given an array A containing prices for different flavors of ice cream, select the two that will cost all the money they have. Print the indices of the flavors of ice-cream that they will choose.

Input Format
Take integer m as input which denotes total money both friends will have. Take integer n as input which denotes total flavors of ice-cream present. Take elements of Array A as an input denoting prices for different flavors of ice cream.

Constraints
It is guaranteed that they will be able to spend all their money
Nested loops are not allowed

Output Format
Print the indices of the flavors of ice-cream that they will choose. In case of multiple pair, print anyone of them.

Sample Input 0
Money - 4
Flavours Available - 5
1 4 5 3 2
Sample Output 0

0 3
Explanation: A[0] + A[3] = 4

Sample Input 1
Money - 4
Flavours Available - 4
2 2 4 3
Sample Output 1
0 1
Explanation: A[0] + A[1] = 4

Question 8

Given an array A containing distinct integers. An integer x is present in the array A, design and implement an algorithm that rearranges the elements of the array such that the following properties hold (after rearrangement):-
All the elements that are smaller than x are placed before x which is followed by all the elements that are larger than x.

Input Format
The first line of the input is the size of the array, followed by the array elements in the next line. The third line gives the element x in the array.

Constraints
You are not allowed to define any additional arrays (other than the given array) for this question.
Note that the final array need not be sorted.

Output Format
All the elements that are smaller than x are placed before x which is followed by all the elements that are larger than x.

Sample Input 0
Size of Array - 7
Array Elements - 77 4 34 12 9 45 1
Element x - 12
Sample Output 0
1 9 4 12 77 45 34

Please note that the solution may not be unique. There may be various correct outputs.

Question 9

You are given an array of integers of size n. Write a function check () that returns true, if and only if the number of occurrences of each value in the array is unique. For all other cases, it must return false.

Input Format
Enter the number of elements in the array and create the array.

Constraints
Note: Print the result in the main() function.

Output Format
true or false.

Sample Input 0
10
1 2 2 3 3 3 4 4 4 4
Sample Output 0
true
Explanation: The value 1 has one occurrence, 2 has two occurrences, 3 has three occurrences and 4 has four. No two values have the same number of occurrences in the array.

Question 10

Suppose you have an array cost[ ] of size n where cost[i] is the cost of the ith step on a staircase. Once you pay the cost, you can either climb one or two steps of the stairs. You can either start from the step with index 0 or step with index 1. Write a function minCost(int cost[ ], int n) that computes and returns an integer value which is the minimum cost that is incurred in reaching the top floor.

Input Format
Enter the number of elements in the array and create the array.

Constraints
Note: Print the result in the main() function.

Output Format
Minimum cost in reaching the top floor.

Sample Input 0
3
10 15 20

Sample Output 0
Minimum Cost = 15
Explanation: Cheapest is: start on cost[1], pay that cost, and go to the top.

Sample Input 1
10
1 100 1 1 1 100 1 1 100 1
Sample Output 1
Minimum Cost = 6
Explanation: Cheapest is: start on cost[0], and only step on 1s, skipping cost[3].
Hint: minCost[i] = cost[i] + min(minCost[i - 1], minCost[i - 2]) for i ¿= 2. minCost[0] = cost[0], minCost[1] = cost[1]

Submission

Please submit your homework in piazza under hw2 folder and make it a private submission to the instructors (not individual TA's or Subidh Sir). Zip all the codes and name the zip as **yourname_rollno**.

The submission deadline is 8:00 pm September 30.